EVT System Carrier
External radar front-end: TI AWR2944PEVM
Carrier contains no 76-81 GHz RF routing

## POWER_INPUT

Файл: 10_POWER_INPUT.kicad_sch

- 3V3
- RADAR_12V
- SYS_EFUSE_FLT
- RADAR_EFUSE_FLT
- GND

## MCU_CORE

Файл: 20_MCU_CORE.kicad_sch

- ETH_MDIO
- ETH_NRST
- IMU_NSS
- IMU_SCK
- IMU_MOSI
- GNSS_RX_EXT
- CAN_TX
- ETH_MDC
- ETH_TX_EN
- ETH_TXD0
- ETH_TXD1
- ETH_NRSTD
- BOOT0
- RADAR_EFUSE_FLT
- SYS_EFUSE_FLT
- ETH_RXD1
- ETH_RXD0
- ETH_CRS_DV
- ETH_50M
- CAN_RX
- GNSS_TX_EXT
- IMU_MISO
- IMU_DRDY
- RADAR_REF
- GNSS_PPS
- 3V3
- GND

## TIME_GNSS

Файл: 30_TIME_GNSS.kicad_sch

- GNSS_PPS
- GNSS_TX_EXT
- GNSS_RX_EXT
- 3V3
- GND

## IMU

Файл: 40_IMU.kicad_sch

- IMU_DRDY
- IMU_MISO
- IMU_MOSI
- IMU_SCK
- IMU_NSS
- 3V3
- GND

## RADAR_IO_CAN

Файл: 50_RADAR_IO_CAN.kicad_sch

- CAN_RX
- RADAR_REF
- RADAR_12V
- CAN_TX
- 3V3
- GND

## ETHERNET

Файл: 60_ETHERNET.kicad_sch

- ETH_MDIO
- ETH_50M
- ETH_CRS_DV
- ETH_RXD0
- ETH_RXD1
- ETH_NRST
- ETH_TXD1
- ETH_TXD0
- ETH_TX_EN
- ETH_MDC
- 3V3
- GND

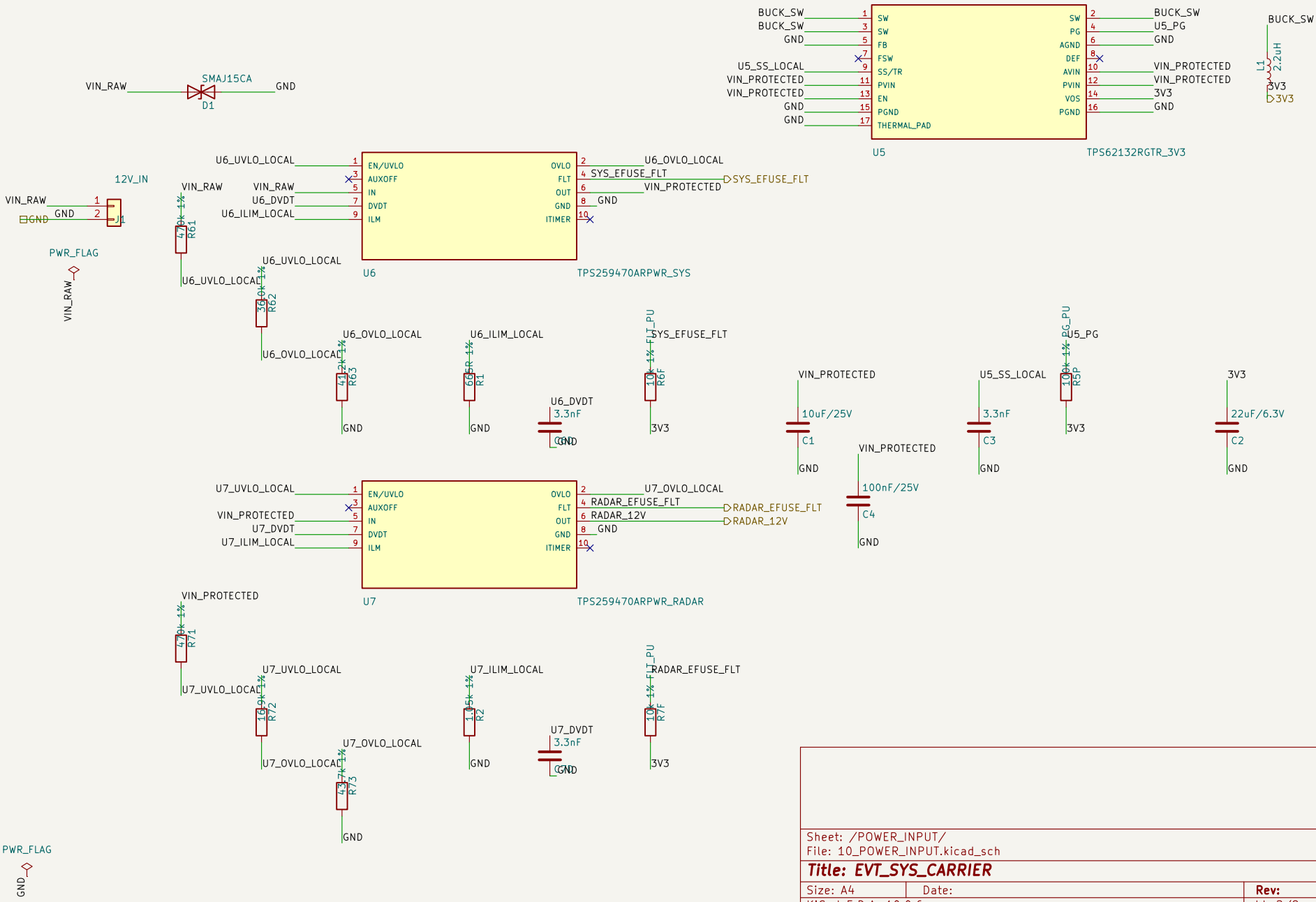
## DEBUG_BOOT

Файл: 70_DEBUG_BOOT.kicad_sch

- NRST
- 3V3
- BOOT0
- GND

| | |
|---|---|
| Sheet: / | |
| File: EVT_SYS_CARRIER.kicad_sch | |
| **Title: EVT_SYS_CARRIER** | |
| Size: A2 | Date: |
| KiCad E.D.A. 10.0.6 | Id: 1/8 |

POWER_INPUT — Rev A authoritative capture
TPS259470A protection + fixed 3.3V TPS62132




| | |
|---|---|
| Sheet: /POWER_INPUT/ | |
| File: 10_POWER_INPUT.kicad_sch | |
| **Title: EVT_SYS_CARRIER** | |
| Size: A4 | Date: |
| KiCad E.D.A. 10.0.6 | Rev: Id: 2/8 |

MCU_CORE — STM32H753IIT6
P0 time capture pins remain fixed by PINMAP_SYS_001.csv

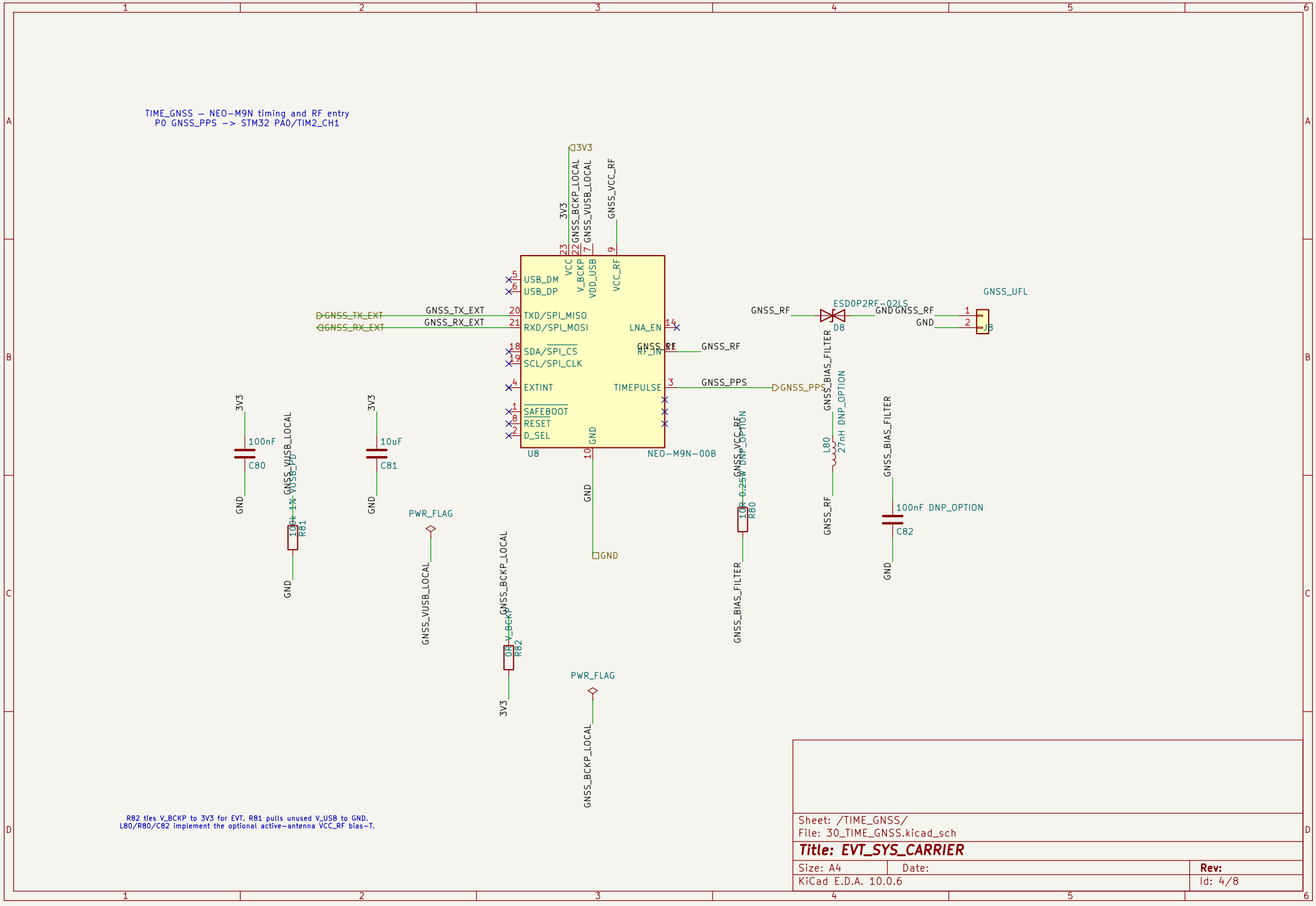
MCU_HSE_25M
TRI_STATE
OUT
GND
VDD
GND
3V3
Y1
ECS-2520S33-250-FN-TR

3V3
FB1
BLM18BD601SH1
VDDA_FILTERED
C15 100nF
C13 100nF
C14 1uF
C10 2.2uF
C11 2.2uF
C12 4.7uF
VCAP1_LOCAL
VCAP2_LOCAL
PWR_FLAG
R10 10k
BOOT0
TC2030_SWD
3V3
SWDIO
GND
SWCLK
GND
SWO
PWR_FLAG
J10
C100 100nF VDD
C101 100nF VDD
C102 100nF VDD
C103 100nF VDD
C104 100nF VDD
C105 100nF VDD
C106 100nF VDD
C107 100nF VDD
C108 100nF VDD
C109 100nF VDD
C110 100nF VDD
C111 100nF VDD
C112 100nF VDD
C113 100nF VDD
C114 100nF VDD33_USB
C115 100nF NRST
DEBUG_TX
TP1
DEBUG_RX
TP2

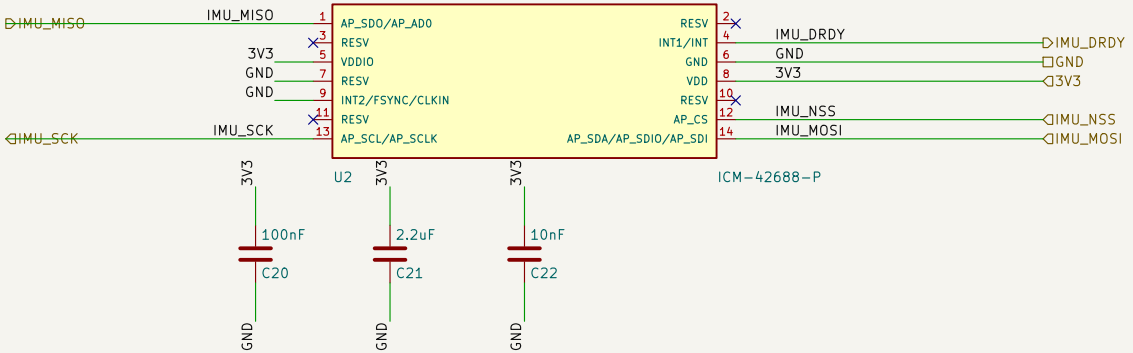
U1 STM32H753IIT6

| | | | |
|---|---|---|---|
| NRST | 31 | NRST | |
| BOOT0 | 166 | BOOT0 | |
| VDDA_FILTERED | 38 | VREF+ | |
| 3V3 | 171 | PDR_ON | |
| | 131 | PI0 | |
| | 132 | PI1 | |
| | 133 | PI2 | |
| | 134 | PI3 | |
| | 173 | PI4 | |
| | 174 | PI5 | |
| | 175 | PI6 | |
| | 176 | PI7 | |
| | 7 | PI8 | |
| | 11 | PI9 | |
| | 12 | PI10 | |
| | 13 | PI11 | |
| MCU_HSE_25M | 29 | PH0 | |
| | 30 | PH1 | |
| | 43 | PH2 | |
| | 44 | PH3 | |
| | 45 | PH4 | |
| | 46 | PH5 | |
| | 83 | PH6 | |
| | 84 | PH7 | |
| | 85 | PH8 | |
| | 86 | PH9 | |
| | 87 | PH10 | |
| | 88 | PH11 | |
| | 89 | PH12 | |
| | 128 | PH13 | |
| | 129 | PH14 | |
| | 130 | PH15 | |
| | 66 | PG0 | |
| | 67 | PG1 | |
| | 106 | PG2 | |
| | 107 | PG3 | |
| | 108 | PG4 | |
| | 109 | PG5 | |
| | 110 | PG6 | |
| | 111 | PG7 | |
| | 112 | PG8 | |
| | 152 | PG9 | |
| | 153 | PG10 | |
| ETH_TX_EN | 154 | PG11 | |
| | 155 | PG12 | |
| ETH_TXD0 | 156 | PG13 | |
| ETH_TXD1 | 157 | PG14 | |
| | 160 | PG15 | |
| | 16 | PF0 | |
| | 17 | PF1 | |
| | 18 | PF2 | |
| | 19 | PF3 | |
| | 20 | PF4 | |
| | 21 | PF5 | |
| | 24 | PF6 | |
| | 25 | PF7 | |
| | 26 | PF8 | |
| | 27 | PF9 | |
| | 28 | PF10 | |
| | 59 | PF11 | |
| | 60 | PF12 | |
| | 63 | PF13 | |
| | 64 | PF14 | |
| | 65 | PF15 | |
| VCAP1_LOCAL | 81 | VCAP | |
| VCAP2_LOCAL | 125 | VCAP | |
| 3V3 | 6 | VBAT | |
| 3V3 | 15 | VDD | |
| 3V3 | 23 | VDD | |
| 3V3 | 36 | VDD | |
| 3V3 | 49 | VDD | |
| 3V3 | 62 | VDD | |
| 3V3 | 72 | VDD | |
| 3V3 | 82 | VDD | |
| 3V3 | 91 | VDD | |
| 3V3 | 103 | VDD | |
| 3V3 | 127 | VDD | |
| 3V3 | 136 | VDD | |
| 3V3 | 149 | VDD | |
| 3V3 | 159 | VDD | |
| 3V3 | 172 | VDD | |
| 3V3 | 114 | VDD | |
| VDDA_FILTERED | 39 | VDDA | |
| GND | 14 | VSS | |
| GND | 37 | VSSA | |
| | 40 | PA0 | GNSS_PPS |
| | 41 | PA1 | ETH_50M |
| | 42 | PA2 | ETH_MDIO |
| | 47 | PA3 | |
| | 50 | PA4 | |
| | 51 | PA5 | |
| | 52 | PA6 | |
| ETH_CRS_DV | 53 | PA7 | ETH_CRS_DV |
| | 119 | PA8 | |
| | 120 | PA9 | |
| | 121 | PA10 | |
| | 122 | PA11 | |
| | 123 | PA12 | |
| SWDIO | 124 | PA13 | |
| SWCLK | 137 | PA14 | |
| | 138 | PA15 | |
| | 56 | PB0 | |
| | 57 | PB1 | |
| | 58 | PB2 | |
| SWO | 161 | PB3 | |
| | 162 | PB4 | |
| | 163 | PB5 | |
| | 164 | PB6 | |
| | 165 | PB7 | |
| | 167 | PB8 | |
| | 168 | PB9 | |
| RADAR_REF | 79 | PB10 | RADAR_REF |
| IMU_DRDY | 80 | PB11 | IMU_DRDY |
| IMU_NSS | 92 | PB12 | IMU_NSS |
| IMU_SCK | 93 | PB13 | IMU_SCK |
| IMU_MISO | 94 | PB14 | IMU_MISO |
| IMU_MOSI | 95 | PB15 | IMU_MOSI |
| | 32 | PC0 | |
| ETH_MDC | 33 | PC1 | ETH_MDC |
| | 34 | PC2_C | |
| | 35 | PC3_C | |
| ETH_RXD0 | 54 | PC4 | ETH_RXD0 |
| ETH_RXD1 | 55 | PC5 | ETH_RXD1 |
| | 115 | PC6 | |
| | 116 | PC7 | |
| | 117 | PC8 | |
| | 118 | PC9 | |
| | 139 | PC10 | |
| | 140 | PC11 | |
| | 141 | PC12 | |
| | 8 | PC13 | |
| | 9 | PC14 | |
| | 10 | PC15 | |
| CAN_RX | 142 | PD0 | CAN_RX |
| CAN_TX | 143 | PD1 | CAN_TX |
| | 144 | PD2 | |
| | 145 | PD3 | |
| | 146 | PD4 | |
| DEBUG_TX | 147 | PD5 | |
| DEBUG_RX | 150 | PD6 | |
| | 151 | PD7 | |
| GNSS_RX_EXT | 96 | PD8 | GNSS_RX_EXT |
| GNSS_TX_EXT | 97 | PD9 | GNSS_TX_EXT |
| | 98 | PD10 | |
| | 99 | PD11 | |
| | 100 | PD12 | |
| | 101 | PD13 | |
| | 104 | PD14 | |
| | 105 | PD15 | |
| | 169 | PE0 | |
| | 170 | PE1 | |
| SYS_EFUSE_FLT | 1 | PE2 | SYS_EFUSE_FLT |
| RADAR_EFUSE_FLT | 2 | PE3 | RADAR_EFUSE_FLT |
| ETH_NRST | 3 | PE4 | ETH_NRST |
| | 4 | PE5 | |
| | 5 | PE6 | |
| | 68 | PE7 | |
| | 69 | PE8 | |
| | 70 | PE9 | |
| | 73 | PE10 | |
| | 74 | PE11 | |
| | 75 | PE12 | |
| | 76 | PE13 | |
| | 77 | PE14 | |
| | 78 | PE15 | |

Sheet: /MCU_CORE/
File: 20_MCU_CORE.kicad_sch
**Title: EVT_SYS_CARRIER**
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 3/8


TIME_GNSS — NEO-M9N timing and RF entry
P0 GNSS_PPS -> STM32 PA0/TIM2_CH1
3V3
GNSS_BCKP_LOCAL
GNSS_VUSB_LOCAL
GNSS_VCC_RF
USB_DM
USB_DP
VCC
V_BCKP
VDD_USB
VCC_RF
GNSS_TX_EXT
GNSS_RX_EXT
TXD/SPI_MISO
RXD/SPI_MOSI
LNA_EN
SDA/SPI_CS
SCL/SPI_CLK
RF_IN
GNSS_RF
EXTINT
TIMEPULSE
GNSS_PPS
SAFEBOOT
RESET
D_SEL
GND
U8
NEO-M9N-00B
ESD0P2RF-02LS
D8
GNSS_UFL
J8
100nF
C80
10uF
C81
100nF DNP_OPTION
C82
10R 0.25W DNP_OPTION
R80
100k 1%
R81
0R
R82
L80
27nH DNP_OPTION
GNSS_BIAS_FILTER
PWR_FLAG
R82 ties V_BCKP to 3V3 for EVT. R81 pulls unused V_USB to GND.
L80/R80/C82 implement the optional active-antenna VCC_RF bias-T.
Sheet: /TIME_GNSS/
File: 30_TIME_GNSS.kicad_sch
Title: EVT_SYS_CARRIER
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 4/8

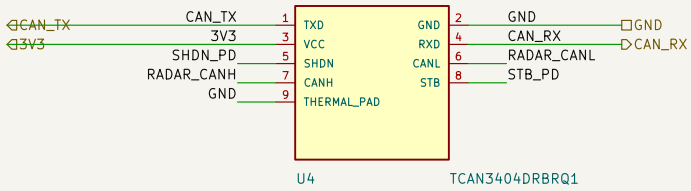
IMU — ICM-42688-P SPI2 + hardware DRDY
P0 IMU_DRDY -> STM32 PB11/TIM2_CH4

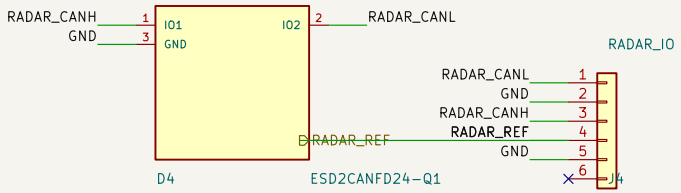


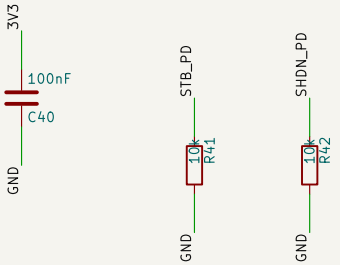

C20/C21 are VDD decoupling; C22 is the dedicated VDDIO 10nF bypass.
Pin7 RESV and pin9 FSYNC are tied to GND per the Rev.A pin contract.

| | | |
|---|---|---|
| Sheet: /IMU/ | | |
| File: 40_IMU.kicad_sch | | |
| **Title: EVT_SYS_CARRIER** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 5/8 |

RADAR_IO_CAN — external AWR2944-class EVM interface
CAN-FD control/metadata + P0 RADAR_REF timestamp + protected 12V

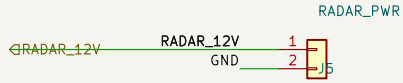


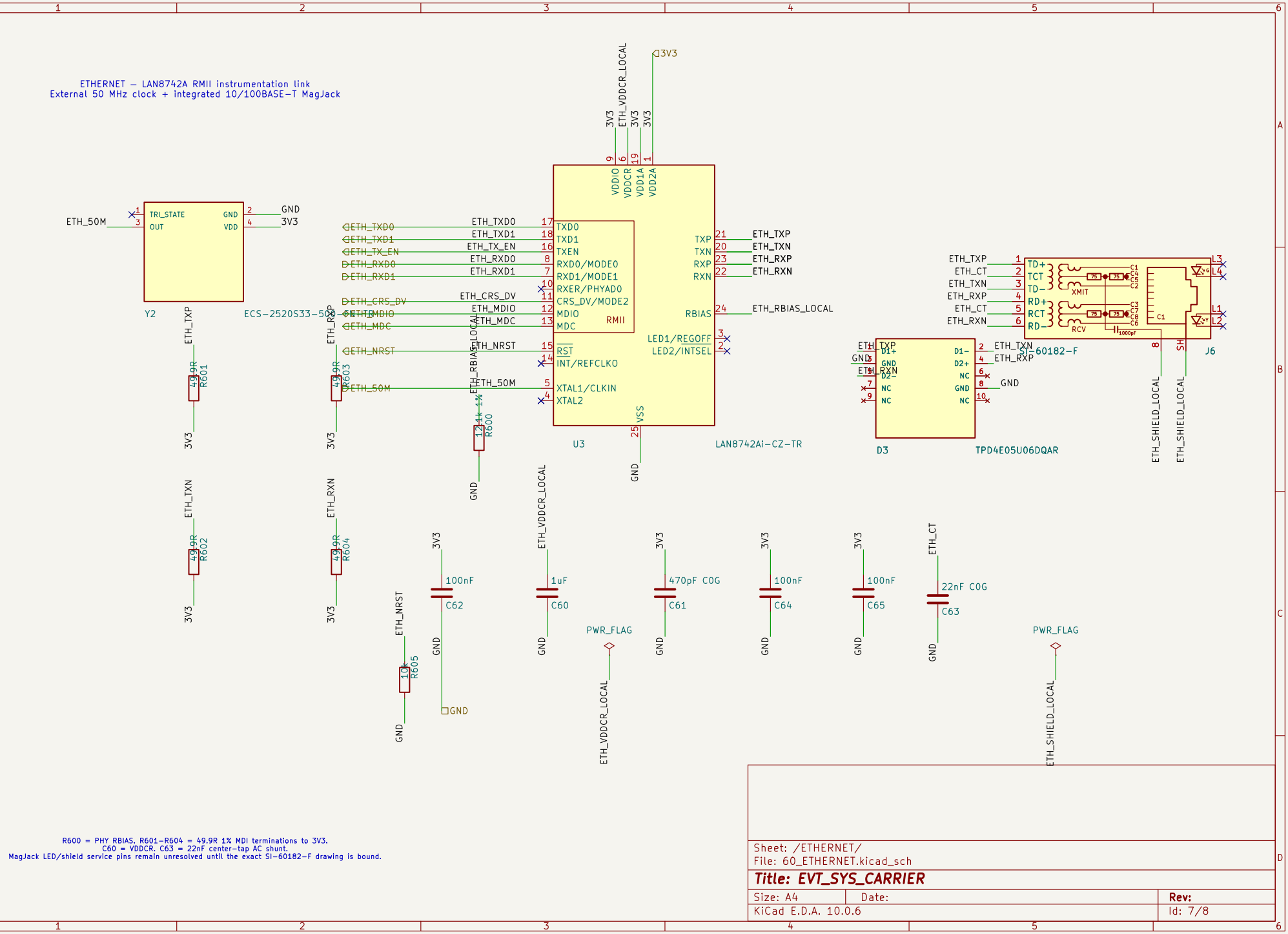






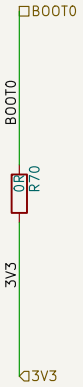

R40 120R 1% is DNP by default; populate only at the CAN endpoint.
R41/R42 10k 1% force normal mode. J4 pin6 remains NC by Rev.A contract.

| | | |
|---|---|---|
| Sheet: /RADAR_IO_CAN/ | | |
| File: 50_RADAR_IO_CAN.kicad_sch | | |
| **Title: EVT_SYS_CARRIER** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 6/8 |


ETHERNET — LAN8742A RMII instrumentation link
External 50 MHz clock + integrated 10/100BASE-T MagJack
ETH_50M
TRI_STATE
OUT
GND
VDD
Y2
ECS-2520S33-50-TR
3V3
ETH_VDDCR_LOCAL
VDDIO
VDDCR
VDD1A
VDD2A
U3
LAN8742Ai-CZ-TR
TXD0
TXD1
TXEN
RXD0/MODE0
RXD1/MODE1
RXER/PHYAD0
CRS_DV/MODE2
MDIO
MDC
RMII
RST
INT/REFCLKO
XTAL1/CLKIN
XTAL2
LED1/REGOFF
LED2/INTSEL
TXP
TXN
RXP
RXN
RBIAS
VSS
ETH_TXD0
ETH_TXD1
ETH_TX_EN
ETH_RXD0
ETH_RXD1
ETH_CRS_DV
ETH_MDIO
ETH_MDC
ETH_NRST
ETH_RBIAS_LOCAL
ETH_TXP
ETH_TXN
ETH_RXP
ETH_RXN
R600
12.1k 1%
R601
R602
R603
R604
49.9R
R605
10k
D3
TPD4E05U06DQAR
J6
60182-F
XMIT
RCV
1000pF
ETH_SHIELD_LOCAL
C60
1uF
C61
470pF C0G
C62
100nF
C63
22nF C0G
C64
100nF
C65
100nF
PWR_FLAG
R600 = PHY RBIAS. R601–R604 = 49.9R 1% MDI terminations to 3V3.
C60 = VDDCR. C63 = 22nF center-tap AC shunt.
MagJack LED/shield service pins remain unresolved until the exact SI-60182-F drawing is bound.
Sheet: /ETHERNET/
File: 60_ETHERNET.kicad_sch
Title: EVT_SYS_CARRIER
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 7/8

DEBUG_BOOT — SWD, reset and boot service access
No vehicle actuation/control interfaces




Primary Tag—Connect SWD connector remains captured with MCU_CORE.
SW1 pulls NRST low; R70 is a DNP service override only.

Sheet: /DEBUG_BOOT/
File: 70_DEBUG_BOOT.kicad_sch

**Title: EVT_SYS_CARRIER**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/8 |